**Problem 1: Finding Complexity using Counter Method**

| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:12 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:25 AM |
| **Time taken** | 12 mins 57 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1**

Correct

Mark 1.00 out of 1.00

Flag question

**Question text**

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
   int i= 1;
   int s =1;
   while(s <= n)
   {
       i++;
       s += i;
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and printf() statements. count variable

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

Answer:(penalty regime: 0 %)

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 9 | 12 | 12 | |
| | 4 | 9 | 9 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.

**Problem 2: Finding Complexity using Counter method**

| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:25 AM |

| | |
|---|---|
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:37 AM |
| **Time taken** | 12 mins 35 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1**

Correct

Mark 1.00 out of 1.00

Flag question

**Question text**

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
   if(n==1)
   {
     printf("*");
   }
   else
   {
    for(int i=1; i<=n; i++)
    {
      for(int j=1; j<=n; j++)
      {
         printf("*");
         printf("*");
         break;
```

```
    }
   }
  }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

Answer:(penalty regime: 0 %)

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 2 | 12 | 12 | |
| | 1000 | 5002 | 5002 | |
| | 143 | 717 | 717 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.

**Problem 3: Finding Complexity using Counter Method**

| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:37 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:41 AM |

| | |
|---|---|
| **Time taken** | 3 mins 32 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1**

Correct

Mark 1.00 out of 1.00

Flag question

**Question text**

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

**Input:**
A positive Integer n
**Output:**
Print the value of the counter variable

Answer:

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 12 | 31 | 31 | |
| | 25 | 54 | 54 | |
| | 4 | 12 | 12 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.

**Problem 4: Finding Complexity using Counter Method**

| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:41 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:44 AM |
| **Time taken** | 2 mins 55 secs |

| | |
|---|---|
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1**

Correct

Mark 1.00 out of 1.00

Flag question

**Question text**

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and printf() statements. count variable

**Input:**
A positive Integer n
**Output:**
Print the value of the counter variable

Answer:

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 4 | 30 | 30 | |
| | 10 | 212 | 212 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.

**Problem 5: Finding Complexity using counter method**

| | |
|---|---|
| **Started on** | Friday, 14 August 2026, 8:45 AM |
| **State** | Finished |
| **Completed on** | Friday, 14 August 2026, 8:49 AM |
| **Time taken** | 3 mins 50 secs |
| **Marks** | 1.00/1.00 |

| Grade | **10.00** out of 10.00 (**100**%) |
|---|---|

**Question 1**

Correct

Mark 1.00 out of 1.00

Flag question

**Question text**

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
  int rev = 0, remainder;
  while (n != 0)
   {
      remainder = n % 10;
      rev = rev * 10 + remainder;
      n/= 10;

   }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and printf() statements. count variable

**Input:**
A positive Integer n
**Output:**
Print the value of the counter variable

Answer:

**Feedback**

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 12 | 11 | 11 | |
| | 1234 | 19 | 19 | |

Passed all tests!

**Correct**

Marks for this submission: 1.00/1.00.